



DeFi Staking Platform — Project Documentation

DeFi Staking Platform

Decentralized staking platform built on **SCAI Mainnet** · Solidity · React · TypeScript · Ethers.js · MetaMask

Live Demo	https://vercel.com/stakingcontract/de-fi-staking-platform
Demo Recording	Loom Walkthrough
GitHub Repo	feudcommon/DeFi-staking-platform
Token Contract (SCAI)	<code>0x4EF03D37c441BcF78D61367f4EE709027632d929</code>
Staking Contract (SCAI)	<code>0x9aab06FAE31e082c26979afca9E53897dB57D50C</code>

1. System Architecture Design

The DeFi Staking Platform allows users to stake ERC20 tokens and earn rewards. It is composed of five main components working in sequence:

#	Component	Role
1	User	Interacts with the application through the frontend
2	MetaMask Wallet	Manages user accounts and signs blockchain transactions
3	React Frontend	Provides the UI for staking, withdrawing, and claiming rewards
4	Staking Smart Contract	Handles staking operations, reward calculations, and withdrawals

#	Component	Role
5	ERC20 Token Contract	Manages token balances and transfers

The frontend communicates with the staking contract using **Ethers.js**. Users connect their wallets through MetaMask and interact with the staking contract to stake tokens, claim rewards, and withdraw funds.

Tech Stack

- Solidity — Smart contract development
- React + TypeScript — Frontend UI
- Ethers.js — Blockchain interaction layer
- MetaMask / WalletConnect / Coinbase Wallet — Wallet connectivity

2. Project Requirements

Contract Requirements

Function	Signature	Description
Stake Tokens	<code>stake(uint amount)</code>	Deposit ERC20 tokens into the staking contract
Withdraw Tokens	<code>withdraw(uint amount)</code>	Withdraw previously staked tokens
Claim Rewards	<code>claimRewards()</code>	Claim accumulated reward tokens

Frontend Requirements

- **Wallet Connection** — MetaMask / WalletConnect / Coinbase Wallet integration
- **Staking Dashboard** — Displays: staked amount, available rewards, APR, APY, wallet balance
- **Staking Form** — Input for depositing tokens
- **Withdraw Form** — Input for withdrawing staked tokens
- **Rewards Section** — View and claim accumulated rewards

3. Use Cases

Use Case 1 — Stake Tokens

Actor: User

1. Connect wallet
2. Enter amount to stake
3. Click Stake
4. Approve transaction in MetaMask
5. Tokens deposited into staking contract

Use Case 2 — View Rewards

Actor: User

1. Open staking dashboard
2. View accumulated reward balance

Use Case 3 — Claim Rewards

Actor: User

1. Click **Claim Rewards**
2. Confirm transaction in MetaMask
3. Receive reward tokens in wallet

Use Case 4 — Withdraw Stake

Actor: User

1. Enter withdrawal amount
 2. Confirm transaction in MetaMask
 3. Receive staked tokens back in wallet
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4. Features Summary

- ✓ Stake tokens
 - ✓ Withdraw tokens
 - ✓ Claim rewards
 - ✓ View dashboard (staked amount, rewards, APR, APY, wallet balance)
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5. Setup & Deployment

Local Setup

```
npm install
npm start
```

Deployment

See [Frontend/DEPLOYMENT.md](#) in the repository for full deployment instructions.

Frontend is deployed on **Vercel**: <https://vercel.com/stakingcontract/de-fi-staking-platform>

Smart Contracts (SCAI Mainnet)

Token Contract: **0x4EF03D37c441BcF78D61367f4EE709027632d929**
Staking Contract: **0x9aab06FAE31e082c26979afca9E53897dB57D50C**

DeFi Staking Platform · Built on SCAI Mainnet